

Cross-synthesis Stein owner and full-internal companion boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope: phase-neutral truth before a preferred pattern

No spatial phase is assumed in this note. BCC is one candidate among a uniform state, other modulated states, metastable states, and failure of the present effective action. The purpose is narrower and falsifiable: determine which signed owner is actually fixed by the registered filtration, and test whether the adverse active-spectator coefficient direction survives when the full internal companions are restored.

Four exact conclusions result.

1. In the registered old owner coordinate, future variance and the adapted R-063 forest are alternate coordinates. They are not additional reserves. The covariance-normal return cancels future variance exactly.
2. For a joint affine Gaussian field displacement and current tangent, the raw square-minus-trace owner depends on their cross synthesis $K = S_x S_v^*$, not on the two marginal covariances alone. Two exact factorizations with identical field and current marginals give opposite owner signs. Endpoint covariance therefore cannot close the production owner.
3. The isolated real radial coefficient plane is a genuine affine saddle. That rules out a proof by global affine coefficient convexity, but it is not a physical control Hessian and not a phase instability.
4. In a separately declared full complex three-component, all-six-row, same-root internal chart, the complete small-noise tensor is strictly positive for every kinetic eigenvalue $a \geq 0$, radius $R \geq 0$, floor $e > 0$, and normalization $P > 0$. Thus the adverse radial sign does not survive in this separately declared internal co-synthesis. Because both the rows and the cross-synthesis rank change, the calculation does not isolate a causal companion and is not yet the production spatial owner. The physical derivative/quadrature cross synthesis and registered spatial incidence remain missing.

The last distinction is load-bearing. “Full internal” is not synonymous with “full production.” No BCC, uniform, or other phase verdict follows.

2. One-use old-owner bookkeeping

Fix one owner o and condition on its strict past $\mathcal{F}_{o-}$. Let

$$b_o = \mathbb{E}[\Phi_o \mid \mathcal{F}_{o-}], \quad \beta_o = \mathbb{E}[\|\Phi_o - b_o\|^2 \mid \mathcal{F}_{o-}], \quad \bar{\tau}_o = \mathbb{E}[\Theta_o \mid \mathcal{F}_{o-}]. \quad (2.1)$$

The conditional complete smaller owner is exactly

$$\boxed{\mathbb{E}[\mathcal{P}_{\text{comp},o} \mid \mathcal{F}_{o-}] = \frac{1}{2}(\beta_o + \|b_o\|^2 - \bar{\tau}_o).} \quad (2.2)$$

The forest bridge is not asserted cell by cell. Impose the complete R-125 hypotheses: common heat, once-owned root covariance, fixed-root recombined visits, exhaustive contraction-closed outputs (or retention of complement leakage), and the established R-063 adapted-forest identification. For the resulting weighted aggregate, R-141 gives

$$\sum_o w_o \mathbb{E} \Delta \mathcal{P}_{\text{comp},o} = \frac{1}{2} \sum_o w_o \mathbb{E} (\Delta F_o^{\text{ad}} - \Delta \mathcal{V}_o). \quad (2.3)$$

The raw current-minus-trace, or covariance-normal, aggregate is

$$\sum_o w_o \mathbb{E} \Delta E_{\text{CN},o} = \sum_o w_o \mathbb{E} (\Delta \mathcal{P}_{\text{comp},o} + \frac{1}{2} \Delta \mathcal{V}_o) = \frac{1}{2} \sum_o w_o \mathbb{E} \Delta F_o^{\text{ad}}. \quad (2.4)$$

Thus the primitive trace occurs with coefficient $-1/2$, the predictable bracket and returned mean each occur with coefficient $+1/2$, future variance cancels on return, and the forest has zero additional multiplicity. Adding forest or future variance again would double-count the same owner.

The complete old-chart relative action retains source and sextic once:

$$\Delta \mathcal{A}_\pi(h) = \sum_\alpha \mathbb{E} \Delta \mathcal{P}_{\text{comp},\alpha} + \frac{9}{20} \|h\|_{\mathcal{S}_\pi}^2 + \frac{3}{20} \mathbb{E} (\|Z_{K,h}\|_6^6 - \|Z_{K,0}\|_6^6). \quad (2.5)$$

Equations (2.3)–(2.5) are complete-aggregate bookkeeping identities under the stated incidence hypotheses, not cellwise identities or estimates.

3. Cross-synthesis Stein owner

Let $g \sim N(0, I_d)$ and let the field and current be affine in the same fresh root:

$$W(g) = w + Xg, \quad V(g) = v + Yg. \quad (3.1)$$

The two syntheses need not agree. Put

$$K = XY^*, \quad \Gamma_x = XX^*, \quad \Gamma_v = YY^*. \quad (3.2)$$

Let $C(W)$ be a smooth finite-dimensional output coefficient and

$$B(W) = C(W)^* C(W), \quad J = C(W)V. \quad (3.3)$$

The primitive trace for the fresh current is

$$\Theta = \text{Tr}\{C(W)\Gamma_v C(W)^*\}. \quad (3.4)$$

Assume that B and its derivatives through order two have sufficient Gaussian integrability to justify the two integrations by parts. Define the raw/covariance-normal diagnostic by

$$2\mathcal{R}_{\text{raw}} = \|J\|^2 - \Theta. \quad (3.5)$$

Theorem 3.1 (exact affine-Gaussian raw owner)

With indices a, b in field space and m, n in current space,

$$\begin{aligned} 2\mathbb{E}\mathcal{R}_{\text{raw}} &= \mathbb{E}(\|J\|^2 - \Theta) \\ &= \mathbb{E}\{v_m B_{mn} v_n + 2v_m K_{an} \partial_a B_{mn} + K_{am} K_{bn} \partial_{ab} B_{mn}\}(W). \end{aligned} \quad (3.6)$$

In particular, if the strict-past current v vanishes,

$$\boxed{2\mathbb{E}\mathcal{R}_{\text{raw}} = \mathbb{E}[K_{am}K_{bn}\partial_{ab}B_{mn}(W)]}. \quad (3.7)$$

Proof. Write $A_i(W) = C(W)Y e_i$. Then

$$\|J\|^2 - \Theta = \|C(W)v\|^2 + 2 \sum_i g_i \langle C(W)v, A_i(W) \rangle + \sum_{i,j} (g_i g_j - \delta_{ij}) \langle A_i(W), A_j(W) \rangle. \quad (3.8)$$

One Gaussian integration by parts in the middle term and two in the last term give (3.6). The contractions are $\sum_i X_{ai}Y_{ni} = K_{an}$. $\square$

For covariance scale ε , replace X, Y by $\sqrt{\varepsilon}X, \sqrt{\varepsilon}Y$. With $v = 0$,

$$\mathbb{E}(\|J\|^2 - \Theta) = \varepsilon^2 \mathbb{E}[K_{am}K_{bn}\partial_{ab}B_{mn}(w + \sqrt{\varepsilon}Xg)]. \quad (3.9)$$

There is no missing factor $1/2$ in (3.9). In a general old-chart endpoint, orthogonality gives

$$\mathcal{R}_{\text{raw}} = \mathcal{P}_{\text{comp}} + \frac{1}{2}\mathcal{V}_{\text{future}}. \quad (3.10)$$

It equals $\mathcal{P}_{\text{comp}}$ only in an explicitly full-reveal chart with $P_f = \text{I}$ and $\mathcal{V}_{\text{future}} = 0$. This distinction is the production firewall, not a notational choice.

4. Endpoint marginals do not identify the owner

The dependence on K in (3.6) is genuine. In two dimensions set

$$B(w_1, w_2) = \begin{pmatrix} (1 + w_1^2)^{-1} & 0 \\ 0 & 1 + w_1^2 \end{pmatrix}. \quad (4.1)$$

This is positive definite for every finite w_1 . Take $X = \text{I}_2$ and compare

$$Y_+ = \text{I}_2, \quad Y_- = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (4.2)$$

Scale both X and $Y_{\pm}$ by $\sqrt{\varepsilon}$, $\varepsilon > 0$. Both choices then have

$$XX^* = Y_+Y_+^* = Y_-Y_-^* = \varepsilon \text{I}_2, \quad (4.3)$$

so their field and current marginal covariances agree. Declare full reveal, so $\mathcal{R}_{\text{raw}} = \mathcal{P}_{\text{comp}}$. If $G \sim N(0, 1)$, direct expectation gives the exact unhalved owners

$$2\mathbb{E}\mathcal{R}_+ = \varepsilon \mathbb{E} \frac{G^2 - 1}{1 + \varepsilon G^2} < 0, \quad 2\mathbb{E}\mathcal{R}_- = 2\varepsilon^2 > 0. \quad (4.4)$$

The first sign is strict because it is $\varepsilon \text{Cov}(G^2, (1 + \varepsilon G^2)^{-1})$ and the second factor is strictly decreasing while G^2 is nonconstant. Equivalently, for independent $X, Y \geq 0$ the symmetrized covariance integrand is

$$(X - Y) \left(\frac{1}{1 + \varepsilon X} - \frac{1}{1 + \varepsilon Y} \right) = - \frac{\varepsilon(X - Y)^2}{(1 + \varepsilon X)(1 + \varepsilon Y)}. \quad (4.5)$$

The corresponding small-noise leading contractions are -2 and $+2$.

Corollary 4.1 (same-marginal sign reversal)

Endpoint field covariance and current covariance, even when both are exactly known, do not determine the full-reveal raw square-minus-trace owner. The joint field-current cross synthesis K is indispensable.

This is a method no-go, not a no-go for the TECT action. It says that the R-146 endpoint covariance normal form cannot by itself determine the R-125 production owner.

5. Radial affine saddle and its boundary

On the real active-spectator coefficient plane, retain the radial column used in R-147:

$$f(x, y) = 4c_0x^2 + 4c_1 \left[x - \alpha \frac{x^2(x - y)}{x^2 + y^2 + e} \right]^2, \quad (5.1)$$

where

$$\alpha = \frac{5}{9}, \quad c_0 = \frac{3}{250P}, \quad c_1 = \frac{243}{8000P}. \quad (5.2)$$

At the diagonal point $(x, y) = (R, R)$, exact differentiation gives

$$H_{11} = \frac{3(117R^4 + 182R^2e + 113e^2)}{1000P(2R^2 + e)^2} > 0, \quad (5.3)$$

$$H_{12} = \frac{3R^2(49R^2 + 27e)}{200P(2R^2 + e)^2} > 0, \quad H_{22} = -\frac{93R^4}{200P(2R^2 + e)^2} < 0, \quad (5.4)$$

and

$$\det H = -\frac{9R^4(3908R^4 + 4718R^2e + 1787e^2)}{50000P^2(2R^2 + e)^4} < 0. \quad (5.5)$$

Thus the affine coefficient energy is a saddle at every $R > 0$. A global affine coefficient-convexity proof is impossible.

This sign is not chart-free. Since $\nabla f(R, R) \neq 0$, a physical lift $z = z(h)$ has

$$D^2(f \circ z) = Dz^*HDz + \nabla f \cdot D^2z. \quad (5.6)$$

The connection term may change the sign. Nor is the active-spectator plane invariant under the registered mass matrix. The saddle therefore cannot be called a physical control Hessian, production instability, or phase verdict.

6. Full complex six-row internal packet

Write $z \in \mathbb{C}^3$ and realify it as

$$u = (\Re z_1, \Re z_2, \Re z_3, \Im z_1, \Im z_2, \Im z_3). \quad (6.1)$$

Let S_A , $A = 1, 2, 3$, be the realifications of the Pauli generators on the first two complex components, extended by zero on the singlet. Define

$$q_A(u) = \frac{u^T S_A u}{|u|^2 + e}, \quad p_A(u) = 2S_A u, \quad \ell_A(u) = 2(S_A - \alpha q_A(u)I_6)u. \quad (6.2)$$

The complete six-row coefficient is

$$C_6(u) = \begin{pmatrix} \sqrt{c_0}p_1^T \\ \sqrt{c_0}p_2^T \\ \sqrt{c_0}p_3^T \\ \sqrt{c_1}\ell_1^T \\ \sqrt{c_1}\ell_2^T \\ \sqrt{c_1}\ell_3^T \end{pmatrix}, \quad B(u) = C_6(u)^T C_6(u). \quad (6.3)$$

The registered internal mass is

$$M = \begin{pmatrix} 1/10 & -1/20 & -1/20 \\ -1/20 & 13/100 & -1/20 \\ -1/20 & -1/20 & 17/100 \end{pmatrix}, \quad C(a) = (aI_3 + M)^{-1}. \quad (6.4)$$

For the declared same-root internal chart put

$$\Gamma(a) = \text{diag}(C(a), C(a)), \quad u_0 = (R, 0, R, 0, 0, 0). \quad (6.5)$$

Here $C(a)$ is the covariance of each real and imaginary coordinate. Denote the A6 complex covariance locally by $\Sigma_{\mathbb{C}}(a) = 2C(a)$; the A7 convention gives $\Gamma = (1/2)\text{realify}(\Sigma_{\mathbb{C}}) = \text{diag}(C, C)$. Dividing (6.5) by two again would double-apply the complex-to-six-real conversion. The outer $1/2$ in the action is a separate factor.

At u_0 , two transverse eigenvalues are

$$4(c_0 + c_1)R^2 = \frac{339R^2}{2000P}, \quad (6.6)$$

two phase directions are exact kernels, and the radial block is

$$4 \begin{pmatrix} R^2[c_0 + c_1(1 - \alpha t)^2] & -c_1 \alpha t(1 - \alpha t)R^2 \\ -c_1 \alpha t(1 - \alpha t)R^2 & c_1 \alpha^2 t^2 R^2 \end{pmatrix}, \quad t = \frac{R^2}{2R^2 + e}. \quad (6.7)$$

These are the R-142 full-fibre checksums.

7. Strict full-internal same-root sign

Let $g \sim N(0, I_6)$ and

$$q = \sqrt{\varepsilon} \Gamma^{1/2} g. \quad (7.1)$$

The declared internal current tangent is the same q . Define the unhalved raw/covariance-normal packet

$$\mathcal{W}_\varepsilon = \mathbb{E}\{q^T B(u_0 + q)q - \varepsilon \text{Tr}[B(u_0 + q)\Gamma]\}. \quad (7.2)$$

By (3.9),

$$\mathcal{W}_\varepsilon = \varepsilon^2 \mathbb{E}[\Gamma_{am} \Gamma_{bn} \partial_{ab} B_{mn}(u_0 + q)]. \quad (7.3)$$

Hence the leading tensor is

$$T(a, R, e, P) = \Gamma_{am} \Gamma_{bn} \partial_{ab} B_{mn}(u_0). \quad (7.4)$$

Put

$$\rho = \frac{R^2}{e}, \quad d_{\text{den}}(a) = 25000a^3 + 10000a^2 + 1115a + 24. \quad (7.5)$$

Theorem 7.1 (full-internal companion positivity)

The exact tensor is

$$T(a, R, e, P) = \frac{3N(a, \rho)}{160P(2\rho + 1)^4 d_{\text{den}}(a)^2}. \quad (7.6)$$

Writing $N = \sum_{j=0}^4 a^j N_j(\rho)$, the coefficient polynomials are

$$\begin{aligned} N_4 &= 34800000000\rho^4 + 329600000000\rho^3 + 513700000000\rho^2 + 299400000000\rho + 67800000000, \\ N_3 &= 41736000000\rho^4 + 212652000000\rho^3 + 297744000000\rho^2 + 168768000000\rho + 38646000000, \\ N_2 &= 15500240000\rho^4 + 52815580000\rho^3 + 64406160000\rho^2 + 34734420000\rho + 8003790000, \\ N_1 &= 2547660000\rho^4 + 6531756000\rho^3 + 6689376000\rho^2 + 3280296000\rho + 738681000, \\ N_0 &= 167968324\rho^4 + 368965068\rho^3 + 323181116\rho^2 + 138377892\rho + 28354299. \end{aligned} \quad (7.7)$$

All twenty-five coefficients are strictly positive. Therefore

$$\boxed{T(a, R, e, P) > 0 \quad (a \geq 0, R \geq 0, e > 0, P > 0).} \quad (7.8)$$

Proof. Differentiate the six-row Gram in (6.3), contract as in (7.4), substitute (6.4), and clear the positive denominator. This gives (7.6)–(7.7). Every monomial in N is nonnegative on $a, \rho \geq 0$, while $N_0(0) > 0$. The denominator is strictly positive. $\square$

The outer $1/2$ in the action makes the leading action coefficient $T/2$. If one instead defines C to mean the complex covariance $\Sigma_{\mathbb{C}} = 2A^{-1}$, the same six-real matrix must be written $(1/2)\text{diag}(C, C)$; the two notations are identical and must not be mixed.

8. Load-bearing companions and the production firewall

The three linear P rows alone give the exact checksum

$$T_P = \frac{9(200000000a^4 + 114000000a^3 + 23610000a^2 + 2179000a + 83641)}{5Pd_{\text{den}}(a)^2} > 0. \quad (8.1)$$

The nonlinear L contribution is not separately positive. At the exact fixture $a = \rho = 1$, $P = 4$,

$$T_P = 0.1171053\dots, \quad T = 0.09753217\dots, \quad T_L = T - T_P < 0. \quad (8.2)$$

Thus retaining all $P + L$ rows is load-bearing; termwise positivity would be a false proof.

For the active-minus-spectator endpoint displacement $v = (1, 0, -1)$, declare the canonical source lift $h = C(a)^{-1/2}v$. Since $C(a)^{-1} = aI + M$, its exact source Hessian is

$$\frac{9}{10}\|h\|^2 = \frac{9}{10}v^T(aI + M)v = \frac{9(200a + 37)}{1000} > 0. \quad (8.3)$$

The R-142 terminal sextic Hessian is also nonnegative. These facts do not close the production owner because (7.6) assumes the special co-synthesis

$$K = \Gamma_x = \Gamma_v = \Gamma. \quad (8.4)$$

For a real production spatial current, the field root and its derivative or quadrature have a joint covariance

$$K = S_x S_v^*. \quad (8.5)$$

It need not equal either marginal and can differ or vanish in real-even Fourier coordinates. The correct tensor is

$$K_{am}K_{bn}\partial_{ab}B_{mn}, \quad (8.6)$$

not the tensor in (7.4) unless (8.4) is proved. Output/spatial projectors, common heat, root/visit incidence, adapted feedback, complement leakage, historical complete low, forest, future variance, balanced response, source lift, and terminal relative secant must also be pinned.

Consequently the adverse radial sign does not survive in this separately declared full co-synthesis. Because both the internal rows and the cross-synthesis rank changed, the calculation does not isolate a causal “repairing companion.” It is a route falsifier, not the R-125/R-141 production sign.

9. The reduced production target

The next calculation should remain in the registered old filtration. For endpoints p, q , compute the polarized predictable kernel

$$\boxed{\mathcal{K}_o[p, q] = \bar{\tau}_o[p, q] - \beta_o[p, q] - \langle b_{p,o}, b_{q,o} \rangle.} \quad (9.1)$$

The relative secant uses $(p, q) = (\Sigma, \Delta)$ and must retain the baseline cross. A finite production cylinder must declare:

1. the joint field/current syntheses S_x, S_v and K ;
2. the once-owned low and root covariances;
3. all six rows, common heat, and exhaustive output/spatial projectors;
4. root/visit incidence and later adapted feedback;
5. complement leakage and historical complete low.

Only then may source and sextic be tested against

$$\eta < \frac{5}{11}, \quad \zeta < \frac{13}{50}. \quad (9.2)$$

Pointwise trace matching, ownerwise sign, and Hessian positivity are stronger sufficient routes, not equivalent statements of T-050.

10. Devil's-advocate review

1. **Objection: The negative radial Hessian proves a physical instability. UPHELD against that inference.** Equation (5.6) has a missing connection term, the radial plane is not invariant under M , and the adverse sign does not survive in the separately declared full co-synthesis. This does not identify which changed companion or synthesis causes the sign.
2. **Objection: Endpoint covariance determines the canonical owner. DISMISSED.** Equations (4.1)–(4.4) preserve both marginals and reverse the sign. The cross synthesis K is independent data.
3. **Objection: Theorem 7.1 is already the production spatial owner. UPHELD against the claim.** It assumes $K = \Gamma$ and omits the derivative/quadrature synthesis and registered owner incidence. The theorem is explicitly scoped as an internal same-root diagnostic.
4. **Objection: Positive T makes the complete action stable. UPHELD against the claim.** The tensor is one local small-noise owner coefficient. Relative source synthesis, terminal secant, heat, future, balanced, low, and spatial terms remain unresolved.
5. **Objection: Forest and future variance can be appended to gain more positivity. DISMISSED.** Equations (2.3)–(2.4) prove that they are alternate coordinates. Appending both is double counting.
6. **Objection: The calculation selects BCC or rejects the present PDE. UPHELD against the claim.** No spatial phase ansatz enters the proof. The result is phase-neutral and cannot decide a PDE replacement.

External review is invited for the realification convention, the two Gaussian integrations by parts, every field/current index in (3.6), the six-row order, the twenty-five positive coefficients, the P/L split, and the production scope firewall.

11. Result footer and reproduction

Dependencies: A1, A6, A7, R-063, R-125, R-141, R-142, R-146, R-147, R-148.

Reproduction command:

```
& E:\Dev\TECT.venv\Scripts\python.exe '
  codes/foundations/a13_classii_cross_synthesis_stein_owner_full_internal_companion_boundary_verify.py
```

Result ID: A13-CLASSII-CROSS-SYNTHESIS-STEIN-OWNER-FULL-INTERNAL-COMPANION-BOUNDARY; Ledger ID: R-149.

Precise statement: exact old-owner aggregate one-use cancellation; exact affine-Gaussian raw/CN cross-synthesis owner; same-marginal sign reversal; exact strict full-internal same-root six-row tensor positivity.

Scope: fixed cutoff and positive floor; registered A1 internal covariance; declared same-root co-synthesis diagnostic; no production spatial transport.

Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.

Expected output: primary and independent PASS; integrated verifier PASS.

Falsification gate: any owner factor, cross-synthesis index, marginal equality, realification, fibre Gram, tensor coefficient, P/L split, hash, PDF, or release check fails.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: no complete production owner sign, physical control Hessian, T-050, A13, Nelson, removal, interacting measure, phase selection, BCC preference or exclusion, PDE replacement, or Sector-A closure.

Proof complete: false; T-050 closed: false; Sector A closed: false.

Next required action: pin an actual old-chart finite production field-current joint synthesis and compute the polarized predictable kernel at (Σ, Δ) , retaining complement and complete-low leakage once.